Physics-Informed Neural Networks for Additive Manufacturing

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Context

Applications: Aerospace, medical implants, automotive industries

Benefits:

- Enables complex shapes and geometries
- Supports rapid prototyping
- Reduces material waste

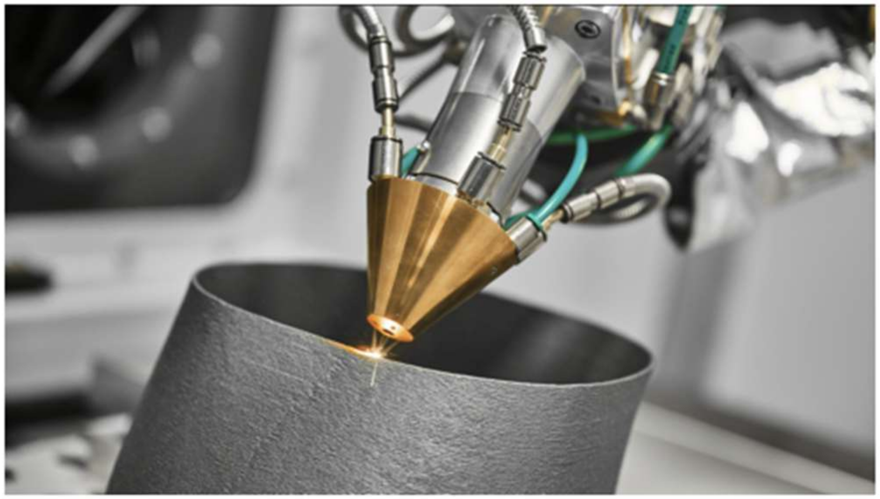


Figure 1 Additive Manufacturing Process

Challenges in Metal Additive Manufacturing

. Quality control is critical in metal AM.

. Key factors affecting quality: temperature, melt pool stability.

. Potential defects from poor control: porosity, cracking, uneven microstructure.

. Accurate Modeling can help prevent these defects

The Melt Pool

. Melt pool is the localized region of molten metal formed by the laser during AM

. Determines microstructure, defects, and overall part quality

. Poor melt pool and temperature control can cause cracks, porosity, and uneven microstructure

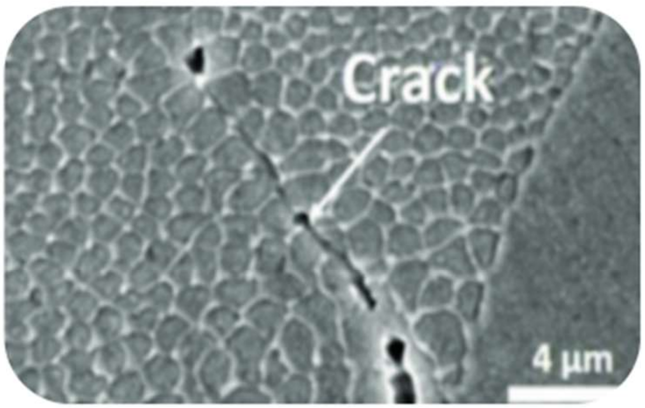


Figure 2 Cracks in microstructures

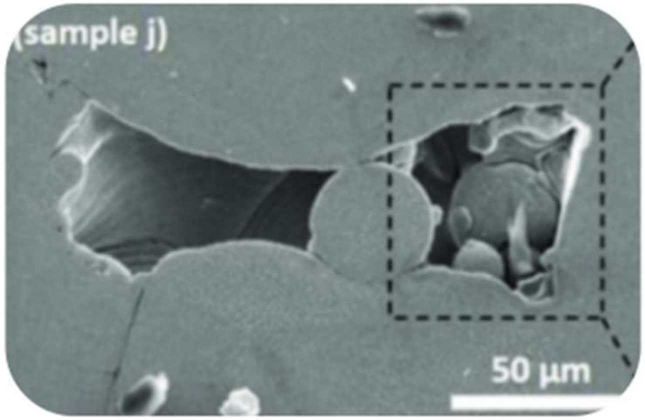


Figure 3 Unmelted material in microstructures

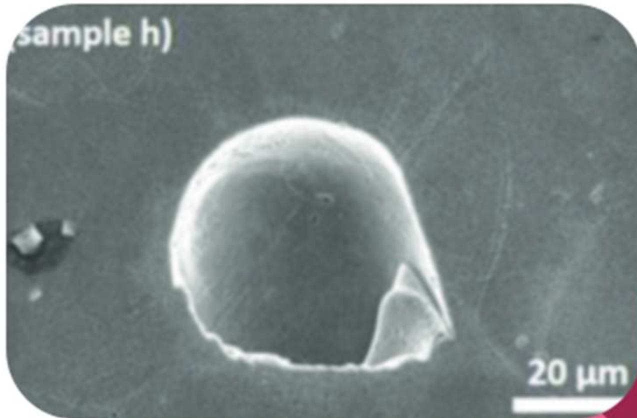


Figure 4 Trapped gas in microstructures

Challenges in Predicting Fluid Behavior at the Melt Pool

. Melt pool is extremely small so small changes strongly affect results

. Very steep temperature gradients exist within a tiny region

Material rapidly changes from liquid to solid

. Flow, heat transfer, and phase change all occur at the same time

. Complexity makes accurate simulation difficult

Current Approaches to Predict Melt Pool Dynamics

. Two main approaches: experiments and numerical simulations

. Experiments are slow and very expensive to perform repeatedly

. High fidelity simulations require significant computational time

. Data driven models depend on large datasets generated from these methods

. Overall: obtaining reliable predictions becomes costly and inefficient

Physics - Informed Neural Networks

. The model learns directly from governing physics, rather that large data sets

. Uses conservation laws such as conservation of energy, mass, and momentum

. Requires far less experimental or simulated training data

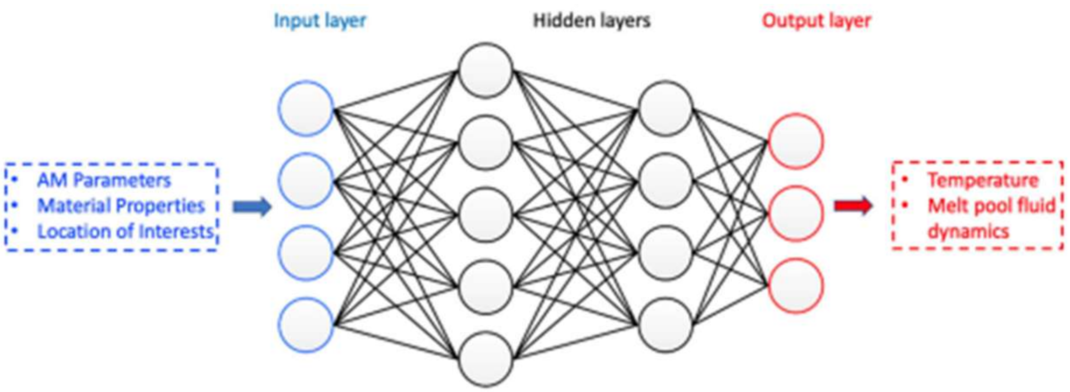
. Goal is accurate predictions while reducing cost and computation time

PINN Framework

Neural Network Approximation:

The PINN approximates the solution fields simultaneously:

$$\hat{\mathbf{u}} = [\hat{T}, \hat{u}_x, \hat{u}_y, \hat{u}_z, \hat{p}] = \mathcal{N}(x, y, z, t;\ \theta)$$



Architecture

5 hidden layers × 250 neurons
Swish activation: σ(x) = x·sigmoid(x)
Fully connected (FCNN)

Inputs → Outputs

[x, y, z, t] → [T, u, p]
Mapping learned via W, b parameters
Optimized by Adam gradient descent

Automatic Differentiation

PDE residuals computed by AD (chain rule)
No truncation/round-off errors
Implemented in TensorFlow

Governing Equations: Energy Conservation

Key Assumptions:

- The solid phase is modeled as a highly viscous fluid with the same constant density as the liquid phase
- Material loss due to vaporization is neglected
- Effects on heat loss, composition change, and fluid motion from evaporation are assumed negligible

Momentum Conservation:

$$\rho(\mathbf{u}_{,t} + \mathbf{u} \cdot \nabla \mathbf{u} - \mathbf{g}) + \nabla p - 2\mu \Delta \mathbf{u} = 0 \qquad (1)$$

Continuity (Mass Conservation):

$$\nabla \cdot \mathbf{u} = 0 \qquad (2)$$

Energy Conservation:

$$(\rho c_p T)_{,t} + \mathbf{u} \cdot \nabla(\rho c_p T) + (\rho L f_L)_{,t} + \mathbf{u} \cdot \nabla(\rho L f_L) - \kappa \nabla^2 T - Q_T = 0 \qquad (3)$$

PINN Loss Function

The total loss combines three components:

$$\mathcal{L} = w_r \mathcal{L}_r + w_b \mathcal{L}_b + w_d \mathcal{L}_d$$

Residual Loss (Lr)

Enforces PDE residual = 0 at collocation points sampled throughout the domain

Boundary Loss (Lb)

Enforces boundary conditions (Dirichlet, Neumann) on domain boundaries

Data Loss (Ld)

Matches PINN predictions to available labeled training data (if any)

Loss Function: Mathematical Formulation

Residual Loss:

$$\mathcal{L}_r = \frac{1}{N_r}\sum_{i=1}^{N_r}|\mathcal{R}(\hat{\mathbf{x}}_i)|^2$$

w_r, w^b, w^d are tunable weights that balance the relative importance of each loss term

Boundary Loss:

$$\mathcal{L}_b = \frac{1}{N_b}\sum_{i=1}^{N_b}|\hat{u}(\mathbf{x}_i) - g(\mathbf{x}_i)|^2$$

Data Loss:

$$\mathcal{L}_d = \frac{1}{N_d}\sum_{i=1}^{N_d}|\hat{u}(\mathbf{x}_i) - u^*(\mathbf{x}_i)|^2$$

Boundary Conditions: Hard vs. Soft

"SOFT" Approach

Method: Add extra penalty terms to loss function for BC

Accuracy: BC satisfaction NOT guaranteed—depends on loss weight

Convergence: Slower—must balance BC weight vs. PDE weight (no theory)

Problem: Choosing optimal λ is empirical and problem-dependent

$$L_{BC} = \Sigma||uNN - u_{bc}||^2 \; at\ boundary\ points$$

⚠ Training loss oscillates; BC accuracy inconsistent (Fig. 4 in paper)

"HARD" Approach

Method: Heaviside function H(x) encodes BC directly in NN output

Accuracy: BC EXACTLY satisfied by construction—no approximation

Convergence: 4–5× faster convergence (Fig. 4, ~5,000 vs ~20,000 iters)

Elegance: No BC weight tuning required—principled formulation

$$\begin{aligned} H(x) &= 1 - \cos[d(x)\pi/\varepsilon] \;\; if\ d(x) < \varepsilon, else\ 1 \\ T_{NN} &= T_{bc}[1 - H(x)] + T \cdot H(x) \end{aligned}$$

✓ Smooth transition; exact BC at boundary; faster training

Network Training & Implementation

- Fully connected neural network with 5 hidden layers and 200 neurons per layer
- Collocation points are randomly sampled across the spatial-temporal domain to enforce PDEs
- Automatic differentiation computes PDE residuals without discretization
- Trained using Adam optimizer followed by L-BFGS for fine-tuning
- No mesh generation required—a key advantage over FEM methods

PINN does NOT replace FEM—it learns from a small FEM-generated training window and generalizes. Think of it as a smart surrogate: FEM provides ground truth; PINN extrapolates efficiently.

Data Generation Workflow

1 **VMS-FEM Solver**

Residual-based variational multi-scale formulation — fully coupled u, p, T

2 **Computational Scale**

4,464,276 tetrahedral elements; $\Delta t = 1\mu s$; 192 CPU cores on Stampede 2 TACC

3 **Validation vs. NIST**

FEM results compared with experimental measurements → validated as 'ground truth'

4 **Selective Sampling**

Only 1.2–1.5 ms window used as labeled training data (small fraction of 0–2.0 ms run)

Temperature & Phase Change Study Setup

Objective:

Validate PINN on a pure thermodynamics problem with phase transition (no fluid flow). This provides insight into PINN's ability to capture energy conservation and phase change.

Problem Setup

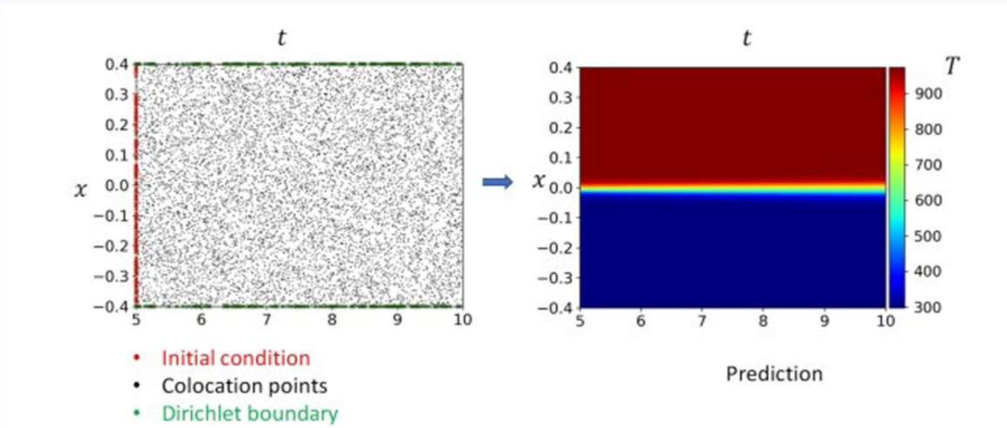
- 2D Stefan melting problem to validate PINN for thermodynamics with phase transition
- Materials: graphite (high thermal conductivity) and aluminum
- Domain: 2D rectangular region with initial solid at temperature below melting point
- Left boundary heated above melting temperature; melting front propagates rightward

PINN Configuration

- 5 hidden layers, 200 neurons per layer
- Informed only with energy conservation (Eq. 3)
- Trained without any labeled data (purely physics-driven)
- Predicts temperature distribution from $t = 5$ s to $t = 10$ s
- Hard Dirichlet BCs applied at left boundary ($T = T_{hot}$)

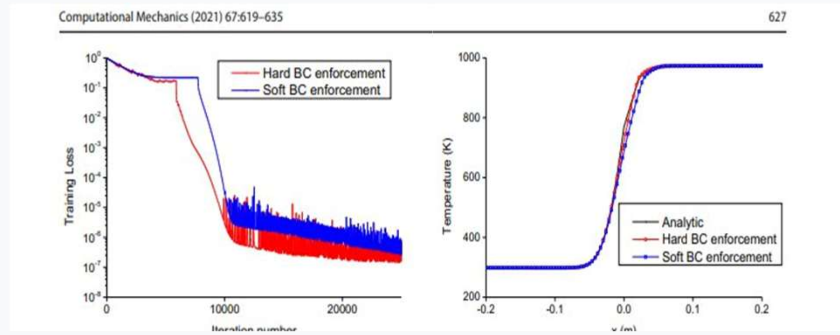
Study 1 Results: PINN Prediction & Boundary Condition Comparison

Fig. 3—PINN Model for Solidification Problem



Left: Collocation points (black), initial conditions (red), Dirichlet BCs (green). Right: Predicted temperature field over x–t domain.

Fig. 4—Hard BC vs. Soft BC Enforcement



Left: Training loss convergence. Right: Temperature at t = 10 s vs. analytical solution.

Key Findings

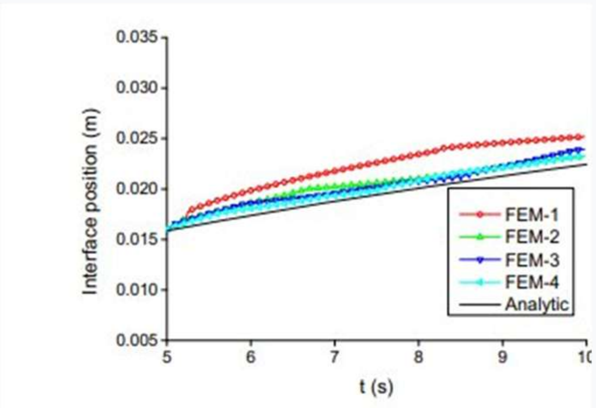
Fig. 3: PINN accurately captures the solidification front propagation through the aluminum domain using only collocation points and energy conservation—no labeled training data.

Fig. 4 (Left): Hard BC enforcement converges several orders of magnitude faster than soft BC, reaching ~10^{-7} training loss.

Fig. 4 (Right): Hard BC produces temperature predictions closely matching the analytical solution; soft BC shows visible deviation near the interface.

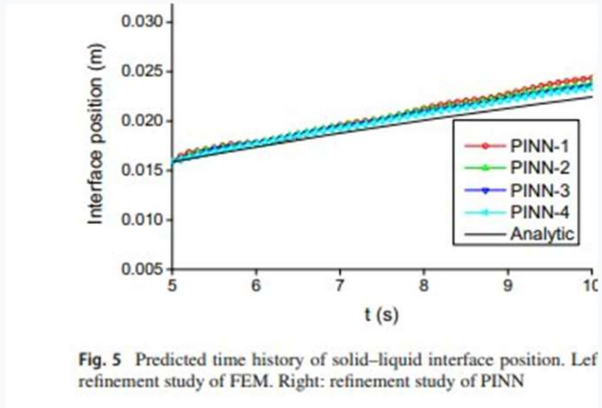
Study 1 Results: PINN vs. FEM Resolution Refinement

Fig. 5-FEM Refinement



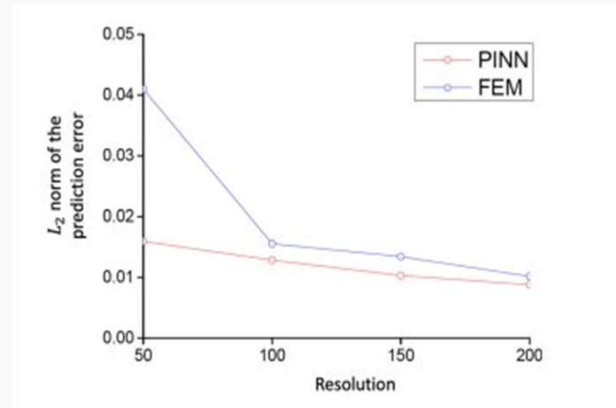
FEM with 4 mesh resolutions vs. analytical

Fig. 5-PINN Refinement



PINN with 4 collocation densities vs. analytical

Fig. 6-L_2 Error Convergence



L_2 prediction error vs. resolution for PINN & FEM

Key Findings

Fig. 5: FEM requires fine meshes (FEM-3, FEM-4) to converge to the analytical interface position. In contrast, all 4 PINN resolutions nearly overlap with the analytical solution—even at only 50 collocation points.

Fig. 6: PINN and FEM achieve similar convergence rates with increasing resolution, but PINN consistently has lower L_2 error, especially at coarse resolutions (N_x = 50: PINN error ~0.015 vs. FEM ~0.042).

Takeaway: Physics-informed loss provides strong regularization, enabling accurate predictions even with sparse collocation points—a significant advantage over mesh-dependent FEM for the Stefan melting problem.

NIST AM-Benchmark Test Series

. PINN framework applied to NIST AM-Bench series

. Public benchmark designed for model verification and comparison

. Model predictions generated without prior access to measured results (blind simulations)

. Provides high-quality experimental data for objective validation

Experimental Setup & Measurements Collected

. Laser Powder Bed Fusion (LPBF) process

. Laser scans across thin powder layer forming a melt pool

. Laser power and scan speed systematically varied

. Experimental measurements collected:
- Temperature fields (in-situ)
- Melt pool geometry (length/width)
- Cooling rates

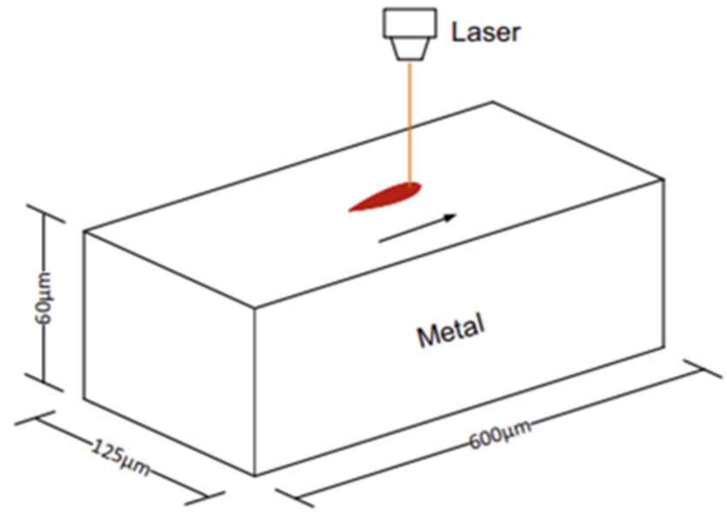


Figure 5 NIST AM Bench test series

Overview of Parameters per Case

. Three laser powder bed fusion experiments with varying laser power and scan speed

. Purpose: Benchmark PINN predictions against experimental measurements

Parameters	Case A	Case B	Case C
Laser power	150 W	195 W	195 W
Scan speed	0.4 m/s	0.8 m/s	1.2 m/s

Table 1 Different laser parameters for each case

Temperature Profile & Distribution Results

. PINN predicts temperature fields that closely follow the experimental profile along the scan track

. Melt pool forms a localized high-temperature region consistent with laser path

. Temperature gradients show rapid cooling away from the melt pool

. Confirms PINN can capture thermal behavior using limited labeled data

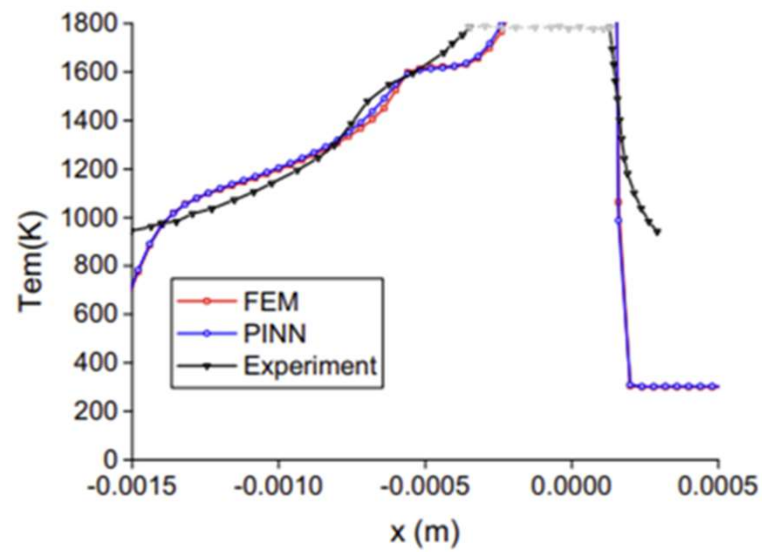


Figure 6 Temperature profile along the scan track for case B at quasi-steady state.

Temperature Profile & Distribution Results

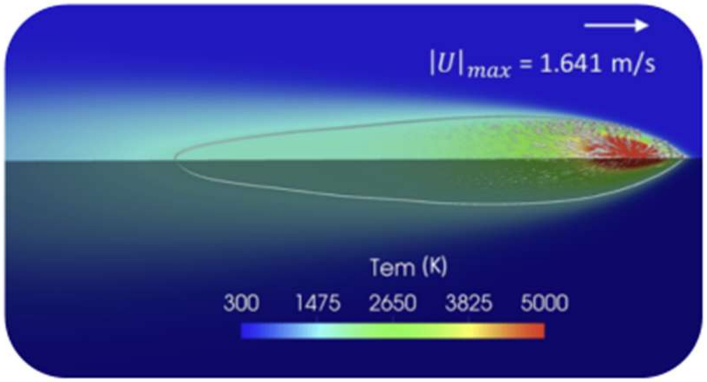


Figure 7 PINN prediction (case A)

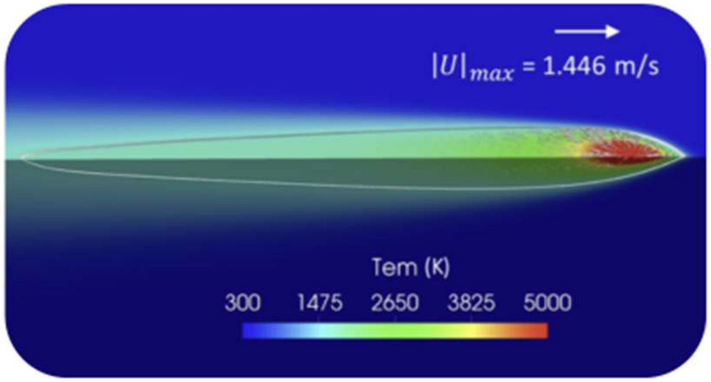


Figure 8 PINN prediction (case C)

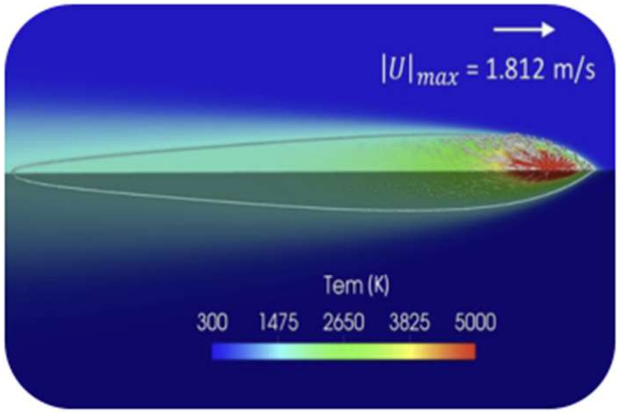


Figure 9 FEM prediction (case B)

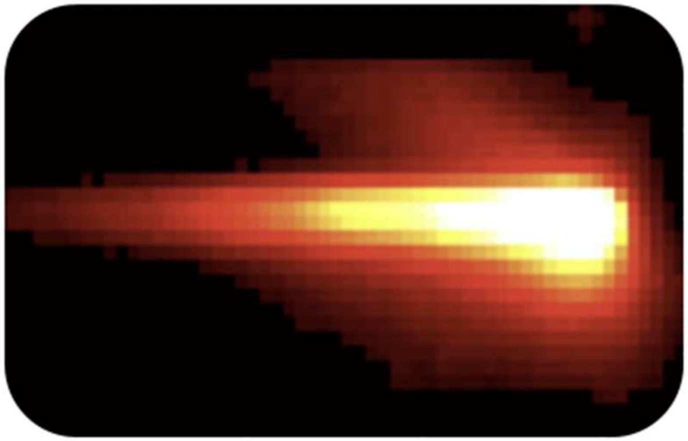


Figure 10 Thermal video frame based on radiance temperature from experiment

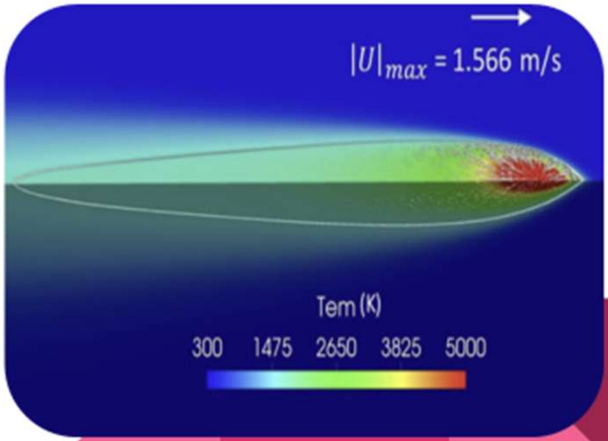


Figure 11 PINN prediction (case B)

Melt Pool Geometry & Dimensions Results

. Melt pool length, width, and depth vary consistently across cases A, B, and C.

- Faster scan speeds yield longer but narrower melt pools
- Lower scan speeds yield shorter but wider/deeper melt pools.

. PINN predictions closely match FEM simulations and NIST experimental measurements.

Cases	Approaches	Length (μm)	Width (μm)	Depth (μm)
Case A	PINN	594.8 (9.7%)	193.3	64.0
	FEM	584.4 (11.3%)	190.8	62.8
	Gan et al. [32]	542 (17.8%)	–	–
	Experiment [119]	659 ± 21	–	–
Case B	PINN	740.3 (5.1%)	160.0	52.8
	FEM	743.6 (4.7%)	157.5	52.5
	Gan et al. [32]	843 (8.1%)	–	–
	Experiment [119]	782 ± 21	–	–
Case C	PINN	732.5 (2.9%)	131.5	43.2
	FEM	727.2 (3.6%)	130.3	42.6
	Gan et al. [32]	785 (4.1%)	–	–
	Experiment [119]	754 ± 46	–	–

Table 2 Melt pool dimensions of case A, B, and C

Melt Pool Geometry & Dimensions PINN Predictions

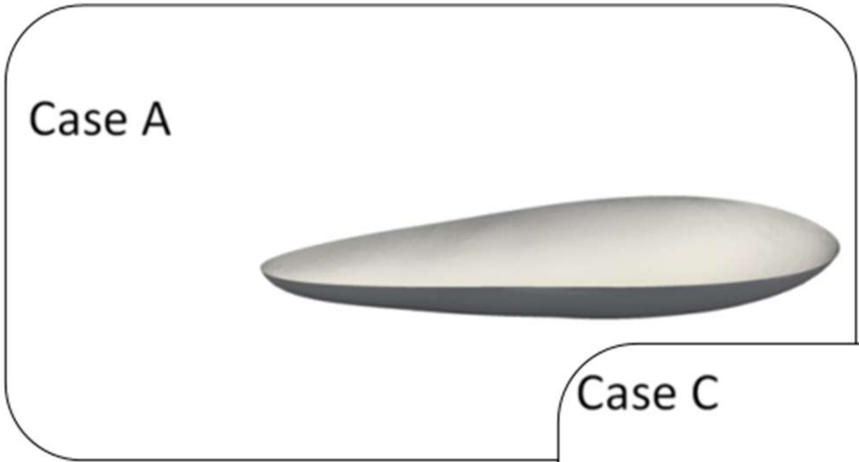


Figure 12 594.8 x 193.3 x 64 (µm)

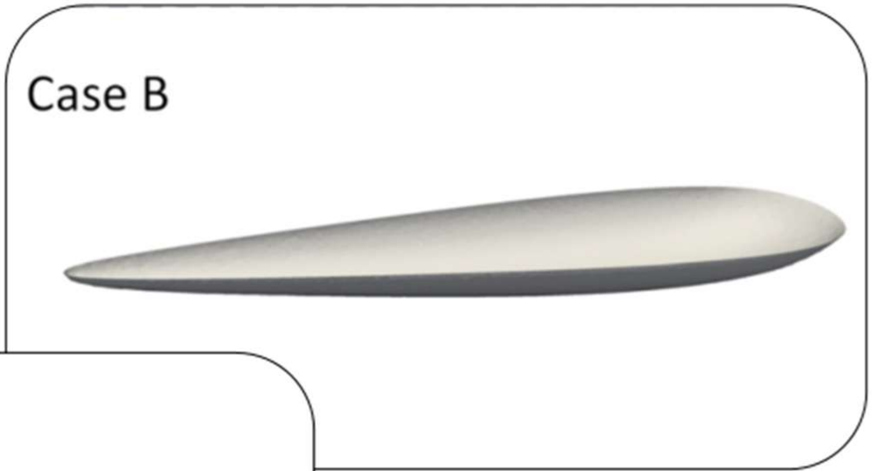


Figure 14 740.3 x 160.0 x 52.8 (µm)

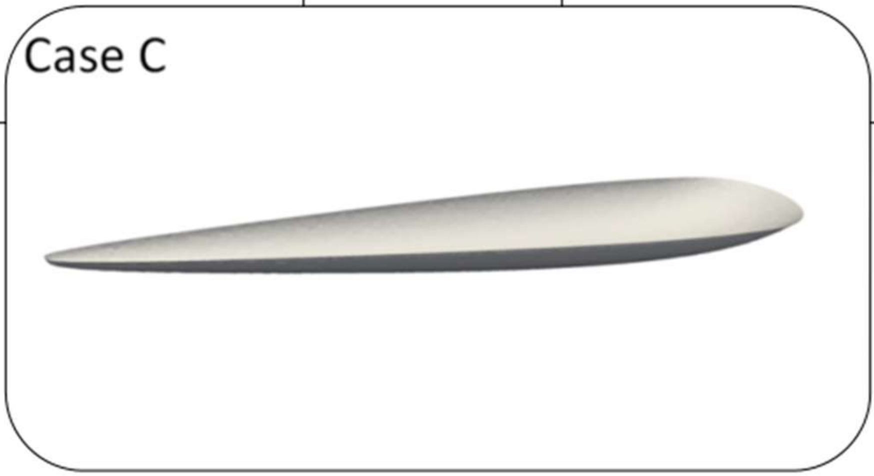


Figure 13 732.5 x 131.5 x 43.2 (µm)

Cooling Rate Results

. PINN predictions show good agreement with FEM and NIST experimental measurements

. Relative discrepancies are larger for cooling rates than for melt pool dimensions (especially Case A)

. Cooling rate increases from Case A to Case C due to higher scan speeds

. Cooling rate is calculated using the solidus temperature and laser scan speed

$$R_c = \frac{T_s - 1273.15K}{t_c} \qquad t_c = (D_s - D_{1273.15})/V_s$$

Cooling Rate Results

Cases	Approaches	Solid cooling rate (K/s)
Case A	PINN	8.54×10^5 (37.7%)
	FEM	8.00×10^5 (29.0%)
	Gan et al. [32]	5.11×10^5 (17.6%)
	Experiment [119]	$6.20 \times 10^5 \pm 7.99 \times 10^4$
Case B	PINN	8.59×10^5 (8.1%)
	FEM	8.16×10^5 (12.7%)
	Gan et al. [32]	6.89×10^5 (26.3%)
	Experiment [119]	$9.35 \times 10^5 \pm 1.43 \times 10^5$
Case C	PINN	1.38×10^6 (7.8%)
	FEM	1.38×10^6 (7.8%)
	Gan et al. [32]	11.30×10^5 (11.7%)
	Experiment [119]	$1.28 \times 10^6 \pm 3.94 \times 10^5$

Table 3 Cooling rates of each case

Summary of Findings

- First application of PINN to predict temperature and melt pool fluid dynamics in metal AM (at time of publication)
- Applied to two representative problems; results show accurate prediction with minimal labeled training data
- Demonstrates SciML potential for complex, multi-physics metal AM processes

Two Main Technical Contributions:

SciML Framework for Metal AM

Physics-informed neural networks predict temperature, pressure, and flow in metal printing without needing huge datasets.

Hard Dirichlet Boundary Approach

Makes the model follow exact boundary values (e.g., surface temperature), speeding up learning and improving accuracy.

Importance and Impact

- PINN shows promise as a complementary tool to conventional FEM, not a replacement
- Enables efficient modeling of complicated AM processes with significantly reduced data requirements
- Initial success suggests broader adoption of PINN in advanced manufacturing simulations

Potential Weaknesses

- Ambient gas phase, free-surface deformation, and evaporation are not modeled
- Computational efficiency comparison with FEM is limited due to different hardware setups (GPU vs CPU)
- Performance depends on availability of training data:
  - More training data yields more efficient and accurate PINN predictions
  - Under scenario with no training data, PINN is ~2× slower than FEM

Future Work & Next Steps

- Model multiphase melt pools and free-surface deformation
- Include evaporation effects for heat loss and fluid motion
- Add powder-scale PDEs (level-set / volume-of-fluid methods) for finer resolution
- Improve computational efficiency and benchmark against FEM on identical hardware

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